4.8 Normal Forms and Adequate Connectives

The proof of completeness wrote each row of a truth table as a list of signed atoms $p_1^v, \dots, p_n^v$ (Section 4.8). Join the signed atoms of one row by $\wedge$, and you get a formula that is true in that row and in no other. Join those formulas, one for each row where a given formula φ is true, by $\vee$, and you get a formula with the same truth table as φ. This section turns that observation into two standard shapes for formulas, the disjunctive and the conjunctive normal form. It then reads validity off the conjunctive shape at a glance, and asks how few connectives are really needed.

4.8.1 Conjunctions and disjunctions of many formulas

Normal forms are long conjunctions and disjunctions. Section 4.5 fixed how to write them, in the paragraph on long conjunctions after Proposition 4.5.1. For formulas $\chi_1, \dots, \chi_k$ with $k \geq 1$, $\chi_1 \wedge \chi_2 \wedge \cdots \wedge \chi_k$ is the conjunction **grouped to the right**, $\chi_1 \wedge (\chi_2 \wedge (\cdots \wedge \chi_k))$, and likewise for $\vee$; for $k = 1$ it is χ_1 itself. That paragraph proved, by induction on the number of terms, how such formulas are evaluated: for every valuation v,

$$\begin{aligned} v(\chi_1 \wedge \cdots \wedge \chi_k) = \mathrm{T} &\iff v(\chi_i) = \mathrm{T} \text{ for every } i = 1, \dots, k, \\ v(\chi_1 \vee \cdots \vee \chi_k) = \mathrm{T} &\iff v(\chi_i) = \mathrm{T} \text{ for some } i = 1, \dots, k. \end{aligned} \tag{4.8.1}$$

We use these two lines constantly below, and we cite them by this display.

So a long conjunction is true when **all** its parts are, and a long disjunction when **at least one** is, just as the words say.

4.8.2 Literals, clauses and the two normal forms

The building blocks are the smallest formulas that still carry a sign.

Definition 4.8.1. Literal

A **literal** is an atom or the negation of an atom.

So p, $\neg p$, q_7 and $\neg q_7$ are literals. The signed atom p^v of Section 4.8 is always a literal. The formulas $\neg\neg p$, $\neg(p \wedge q)$ and $\bot$ are not: in the first two, $\neg$ is applied to something other than an atom, and $\bot$ is neither an atom nor a negation. The **opposite** of a literal is $\neg p$ if the literal is p, and p if it is $\neg p$. By the clause for $\neg$ in Definition 4.4.2, a literal and its opposite take opposite values under every valuation.

From literals we build two kinds of blocks and two kinds of formulas.

- A **clause** joins one or more literals by $\vee$, such as $p \vee \neg q \vee r$, or $\neg q$ alone.
- A **conjunct** (in this section, short for "conjunctive clause") joins one or more literals by $\wedge$, such as $p \wedge \neg q \wedge r$, or $\neg q$ alone.

A normal form is a formula whose connectives come in fixed layers: negations only on atoms,

then one layer of one connective, then one layer of the other.

Definition 4.8.2. Disjunctive Normal Form

A formula **has disjunctive normal form** (DNF), or is **in DNF**, when it has the shape $C_1 \vee \cdots \vee C_k$, with $k \geq 1$, where every C_i joins **one or more literals** by $\wedge$.

Definition 4.8.3. Conjunctive Normal Form

A formula **has conjunctive normal form** (CNF), or is **in CNF**, when it has the shape $K_1 \wedge \cdots \wedge K_k$, with $k \geq 1$, where every K_i joins **one or more literals** by $\vee$.

In words: a DNF is an "or of ands" and a CNF an "and of ors", and in both the only negations sit directly on atoms. Read a DNF as a list of situations, any one of which makes it true; read a CNF as a list of requirements, all of which must be met.

Check some formulas against the definitions, clause by clause.

1. $(p \wedge \neg q) \vee (\neg p \wedge q)$ is in DNF: two conjuncts, each a conjunction of literals. It is not in CNF: as a conjunction of clauses it would have to be a single clause (it is not a conjunction), and a clause cannot contain $\wedge$.
2. $(p \vee q) \wedge (\neg p \vee \neg q)$ is in CNF, with two clauses, and not in DNF.
3. $p \wedge \neg q$ is in both: in DNF with one conjunct, and in CNF with two clauses of one literal each. Likewise $p \vee \neg q$ is in both.
4. A single literal, $\neg r$, is in both forms: one conjunct of one literal, and one clause of one literal. This degenerate case matters, because a formula such as $\neg r$ must count as already normal.

A nonexample by a minimal change: $(p \wedge \neg q) \vee r$ is in DNF, but $\neg(p \wedge q) \vee r$ is not in either form. Only one symbol moved, and the literal condition fails: $\neg$ now stands on the conjunction $p \wedge q$, not on an atom. The formula $p \to q$ is in neither form either, since $\to$ is not allowed at all.

Why these layers? Because they are the shapes that the truth table hands us, as the next theorem shows, and because in these shapes some questions become easy: the subsection "Reading validity off a CNF" below decides validity of a CNF without a table.

Warning. A literal is an atom with **at most one** $\neg$, and nothing else. $\neg\neg p$ is equivalent to the literal p but is not a literal, so $\neg\neg p \vee q$ is not in CNF as written. Likewise $\bot$ is not a literal, so a "normal form" containing $\bot$ is not one; that is why the constructions below never use it.

Quick check. Which of the following are in CNF, and which are in DNF? (i) $(p \vee \neg q) \wedge r$; (ii) $\neg(p \vee q)$; (iii) $p \vee (\neg q \wedge r)$.

Solution. (i) CNF, with clauses $p \vee \neg q$ and r; not DNF, since the conjunct would have to be the whole formula and $p \vee \neg q$ is not a literal. (ii) Neither: $\neg$ stands on a disjunction. (iii) DNF, with conjuncts p and $\neg q \wedge r$; not CNF, since the one clause would contain $\wedge$.

4.8.3 Reading a DNF off the truth table

Take a valuation v and atoms $p_1, \dots, p_n$ with $n \geq 1$, and form the conjunct of the signed atoms of the row of v,

$$C_v := p_1^v \wedge \cdots \wedge p_n^v.$$

For every valuation w, each p_i^v is true under w exactly when $w(p_i) = v(p_i)$: if $v(p_i) = \mathrm{T}$, then $p_i^v = p_i$; if $v(p_i) = \mathrm{F}$, then $p_i^v = \neg p_i$, which is true under w exactly when $w(p_i) = \mathrm{F}$. By Eq. 4.8.1,

$$w(C_v) = \mathrm{T} \iff w(p_i) = v(p_i) \text{ for every } i = 1, \dots, n. \tag{4.8.2}$$

So C_v is true in the row of v and in no other row. It is a formula that "names" one row.

Example. Exactly one of two

Find a formula in DNF that is true exactly when exactly one of p, q is true, and check it.

Solution. The rows with exactly one of p, q true are $(p, q) = (\mathrm{T}, \mathrm{F})$ and (F, T). Their conjuncts are $p \wedge \neg q$ and $\neg p \wedge q$, so the answer is

$$(p \wedge \neg q) \vee (\neg p \wedge q).$$

Check against the table: in the row T, F the first conjunct is true; in F, T the second is; in T, T each conjunct has a false literal ($\neg q$, $\neg p$); in F, F likewise (p, q). So the formula is true in exactly the two rows wanted.

The same recipe works for any formula.

Theorem 4.8.1. Disjunctive Normal Form

Every formula φ is equivalent to a formula in DNF. If φ has at least one atom, the DNF can be chosen so that every atom in it is an atom of φ.

Idea. List the rows where φ is true and disjoin their conjuncts C_v. A valuation makes the disjunction true exactly when it agrees with one of those rows, which by Lemma 4.4.1 is exactly when it makes φ true. Two degenerate cases need a convention: a formula with no T-row, whose disjunction would be empty, and a formula with no atoms, which has no rows to speak of.

Proof. By the paraphrase of Definition 4.5.4, two formulas φ and ψ are equivalent exactly when $v(\varphi) = v(\psi)$ for every valuation v.

Case 1: φ has at least one atom. Let $p_1, \dots, p_n$ ($n \geq 1$) be the atoms of φ. A **row** is a choice of a value for each p_i; there are 2^n rows, and for each we fix a valuation that takes those values at $p_1, \dots, p_n$ (and, say, T at every other atom). By Lemma 4.4.1, $v(\varphi)$ depends only on the row of v. Let $v_1, \dots, v_k$ be the fixed valuations of the rows in which φ is T.

If $k \geq 1$, let $D \coloneqq C_{v_1} \vee \cdots \vee C_{v_k}$, which is in DNF by Definition 4.8.2, with atoms among $p_1, \ldots, p_n$. Let w be any valuation. By Eq. 4.8.1 and Eq. 4.8.2, $w(D) = \mathrm{T}$ exactly when w agrees with some v_j on $p_1, \ldots, p_n$. If it does, then $w(\varphi) = v_j(\varphi) = \mathrm{T}$ by Lemma 4.4.1. Conversely, if $w(\varphi) = \mathrm{T}$, then the row of w is one where φ is T, so w agrees on $p_1, \ldots, p_n$ with the valuation v_j fixed for that row. Hence $w(D) = w(\varphi)$ for every w, and $\varphi \equiv D$.

If $k = 0$, then φ is F in every row, hence under every valuation. The formula $p_1 \wedge \neg p_1$ is in DNF (one conjunct of two literals) and is F under every valuation by Definition 4.4.2. So $\varphi \equiv p_1 \wedge \neg p_1$.

Case 2: φ has no atoms. Any two valuations agree on all the atoms of φ (there are none), so by Lemma 4.4.1 $v(\varphi)$ is the same for every v. Let p be an atom. If that value is T, then $\varphi \equiv p \vee \neg p$, which is true under every valuation and is in DNF (two conjuncts of one literal); if it is F, then $\varphi \equiv p \wedge \neg p$. This proves the theorem.

■

The conjuncts C_v are the signed atoms of Section 4.8 joined by $\wedge$. There the signed atoms of a row were premises from which φ^v was derived; here they are joined into a formula that is true in that row only. The proof is the proof of completeness with the derivations taken out.

For $p \wedge \neg p$ we could have written $\bot$, which is also F under every valuation. We did not, because $\bot$ is not a literal, and because the adequacy theorem at the end of this section reuses the construction in a language without $\bot$.

4.8.4 Conjunctive normal form

A DNF lists the rows where a formula is true. A CNF does the opposite: it lists the rows where the formula is **false**, one clause per row, each clause being false in its row and true in every other.

Theorem 4.8.2. Conjunctive Normal Form

Every formula φ is equivalent to a formula in CNF. If φ has at least one atom, the CNF can be chosen so that every atom in it is an atom of φ.

Idea. Apply Theorem 4.8.1 to $\neg\varphi$ and negate the result: $\varphi \equiv \neg\neg\varphi \equiv \neg(C_1 \vee \cdots \vee C_k)$. Now push the $\neg$ inwards with De Morgan's laws and double negation (Proposition 4.5.1): $\neg(C_1 \vee \cdots \vee C_k)$ becomes $\neg C_1 \wedge \cdots \wedge \neg C_k$, and each $\neg C_i$, the negation of a conjunction of literals, becomes the disjunction of the opposite literals. The proof checks the outcome directly with Eq. 4.8.1, which avoids De Morgan's laws for many terms.

Proof. By Theorem 4.8.1, $\neg\varphi \equiv C_1 \vee \cdots \vee C_k$ for some conjuncts $C_1, \ldots, C_k$ (with atoms among those of φ if it has any, since $\neg\varphi$ has the same atoms). For each i, let K_i be the disjunction of the opposites of the literals of C_i, in the same order, and let $K \coloneqq K_1 \wedge \cdots \wedge K_k$. Then K is in CNF, with the same atoms as $C_1 \vee \cdots \vee C_k$.

Let w be a valuation. Each literal of K_i is true under w exactly when the corresponding

literal of C_i is false. By Eq. 4.8.1, used three times:

$$\begin{aligned} w(K) = \mathrm{T} &\iff \text{for every } i,\ w(K_i) = \mathrm{T} \\ &\iff \text{for every } i,\ \text{some literal of } C_i \text{ is false under } w \\ &\iff \text{for every } i,\ w(C_i) = \mathrm{F} \\ &\iff w(C_1 \vee \cdots \vee C_k) = \mathrm{F}. \end{aligned}$$

Since $C_1 \vee \cdots \vee C_k \equiv \neg\varphi$, the last line holds exactly when $w(\neg\varphi) = \mathrm{F}$, that is, when $w(\varphi) = \mathrm{T}$. So $w(K) = w(\varphi)$ for every w, and $\varphi \equiv K$.

■

Unwinding the construction: the conjuncts C_i of the DNF of $\neg\varphi$ are the rows where φ is **false**, and K_i is the clause that is false exactly in the row of C_i. So the CNF can be read straight off the table: **one clause per F-row**, built from the opposites of that row's signed atoms. In the row $p = \mathrm{T}$, $q = \mathrm{F}$, for instance, the clause is $\neg p \vee q$. A formula with no F-row gets, by the proof, the clause of a contradiction's DNF negated, $\neg p \vee p$ (from $C_1 = p \wedge \neg p$), which is true in every row.

Alternatively, a CNF can be reached by **rewriting**. First replace each $\bot$, which is not a literal, by $p \wedge \neg p$ for an atom p; both are false in every row. Replace $\alpha \leftrightarrow \beta$ by $(\alpha \to \beta) \wedge (\beta \to \alpha)$ and $\alpha \to \beta$ by $\neg\alpha \vee \beta$; push each $\neg$ inwards by De Morgan's laws, and remove $\neg\neg$; finally distribute $\vee$ over $\wedge$, replacing $\alpha \vee (\beta \wedge \gamma)$ by $(\alpha \vee \beta) \wedge (\alpha \vee \gamma)$. Each step replaces a part of the formula by an equivalent one: the equivalences are $\bot \equiv p \wedge \neg p$ and those of Proposition 4.5.1, which hold for arbitrary formulas, and Lemma 4.5.2 says the whole formula stays equivalent; since $\equiv$ is transitive (Section 4.5), the steps chain. Stopping before the distribution step, and distributing $\wedge$ over $\vee$ instead, gives a DNF. The rewriting route is often much shorter than the table, as the next example shows, but it can also be much longer, as the warning after it shows.

Example. Two routes to a normal form

Put $\varphi = (p \to q) \to (\neg q \to r)$ into DNF and into CNF, first from its truth table and then by rewriting.

Solution. The table route. Work backwards from F (Section 4.4): φ is false exactly when $p \to q$ is T and $\neg q \to r$ is F. The latter needs $\neg q$ true and r false, so $q = r = \mathrm{F}$; then $p \to q$ true needs $p = \mathrm{F}$. So φ has exactly one F-row, $p = q = r = \mathrm{F}$, as the table confirms:

p	q	r	$p \to q$	$\neg q \to r$	$(p \to q) \to (\neg q \to r)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	F

The DNF of Theorem 4.8.1 has one conjunct for each of the seven T-rows:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \\ \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r).$$

The CNF of Theorem 4.8.2 has one clause for the single F-row, built from the opposites of $\neg p, \neg q, \neg r$: it is the single clause $p \vee q \vee r$.

The rewriting route. Eliminate the three implications, then push the negation in:

$$\begin{aligned} \varphi &\equiv \neg(\neg p \vee q) \vee (\neg\neg q \vee r) \\ &\equiv (\neg\neg p \wedge \neg q) \vee (q \vee r) \\ &\equiv (p \wedge \neg q) \vee q \vee r, \end{aligned}$$

using (c) of Proposition 4.5.1 three times, then De Morgan's law (b) for $\neg(\alpha \vee \beta)$, then double negation (d) twice. This is already a DNF, with three conjuncts. For a CNF, distribute, by (g) together with commutativity (f) and associativity (h): $(p \wedge \neg q) \vee (q \vee r) \equiv (p \vee q \vee r) \wedge (\neg q \vee q \vee r)$. The second clause contains q and $\neg q$, so it is true in every row, and a conjunction with a part that is true in every row is equivalent to its other part. So $\varphi \equiv p \vee q \vee r$: one clause, as the table route found.

The formula $p \vee q \vee r$ is also a DNF, with three conjuncts of one literal each. So the table's seven conjuncts and the rewriting's three can both be replaced by three one-literal conjuncts. Both routes are correct; they differ in length, and neither is guaranteed to give the shortest normal form.

Warning. A formula has **many** equivalent CNFs and DNFs; "normal form" does not mean "unique form". $p \vee q \vee r$ and $(p \vee q \vee r) \wedge (\neg q \vee q \vee r)$ are both CNFs of the formula above. And the two routes have opposite weaknesses. The table route needs all 2^n rows. The rewriting route can explode at the distribution step: distributing $(p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee (p_3 \wedge q_3)$ gives 8 clauses, one for each way of picking p_i or q_i from every conjunct, and n such conjuncts give 2^n clauses.

4.8.5 Reading validity off a CNF

A CNF is a list of requirements, all of which must hold. That makes one question easy.

Proposition 4.8.3. Validity of a CNF

(a) A formula in CNF is a tautology exactly when **every** clause contains some atom together with its negation.

(b) A formula in DNF is satisfiable exactly when **some** conjunct contains no atom together with its negation.

Try it first (ten minutes): take a clause in which no atom occurs both with and without $\neg$, such as $p \vee \neg q \vee r$. Can you find a row that makes it false? What does that row do to a CNF containing this clause?

Idea. The direction "every clause has a pair ⇒ tautology" is the easy one: $p \vee \neg p$ is true in every row, so a clause containing both is too, and a conjunction of clauses true in every row is true in every row. The attempt above is the other direction. To make a CNF false, it is enough to make **one** clause false, and a clause is false only when **all** its literals are. So set every atom of the clause against its literal: F for a plain atom, T for a negated one. The one thing that could stop us is an atom that appears both plain and negated, since it would have to be both F and T. That is exactly the pair the proposition talks about.

Proof. *(a)* Let $K = K_1 \wedge \cdots \wedge K_k$ be in CNF.

($\Leftarrow$) Suppose every clause K_i contains some atom p and its negation $\neg p$. Let v be a valuation. One of p, $\neg p$ is true under v, by Definition 4.4.2, so $v(K_i) = \mathrm{T}$ by Eq. 4.8.1. This holds for every i, so $v(K) = \mathrm{T}$ by Eq. 4.8.1. As v was arbitrary, K is a tautology (Definition 4.5.1).

($\Rightarrow$) We prove the contrapositive. Suppose some clause K_i contains no atom together with its negation. Define a valuation v by: $v(p) := \mathrm{F}$ if the literal p occurs in K_i; $v(p) := \mathrm{T}$ if the literal $\neg p$ occurs in K_i; $v(p) := \mathrm{T}$ for every other atom. This is well defined, because no atom occurs in K_i both as p and as $\neg p$. Every literal of K_i is false under v, so $v(K_i) = \mathrm{F}$ by Eq. 4.8.1, and hence $v(K) = \mathrm{F}$, again by Eq. 4.8.1. So K is not a tautology.

(b) Let $D = C_1 \vee \cdots \vee C_k$ be in DNF.

($\Leftarrow$) Suppose C_i contains no atom together with its negation. Define v by $v(p) := \mathrm{T}$ if p occurs in C_i, $v(p) := \mathrm{F}$ if $\neg p$ occurs in C_i, and $v(p) := \mathrm{T}$ otherwise; this is well defined as in (a). Every literal of C_i is true under v, so $v(C_i) = \mathrm{T}$ and $v(D) = \mathrm{T}$ by Eq. 4.8.1. So D is satisfiable (Definition 4.5.2).

($\Rightarrow$) If every conjunct contains some p and $\neg p$, then under every valuation one of these two literals is false, so every conjunct is false and so is D, by Eq. 4.8.1. So D is not satisfiable. This proves the proposition.

■

The two parts are two faces of one fact. Section 4.5 showed that a formula is a tautology exactly when its negation is unsatisfiable (after Definition 4.5.2). Let D be in DNF, and let K be the CNF whose clauses are the conjuncts of D with every literal replaced by its opposite. The proof of Theorem 4.8.2, applied to $\varphi = \neg D$ with D itself as the DNF of $\neg\varphi \equiv D$, shows $K \equiv \neg D$. A clause of K has an atom together with its negation exactly when the corresponding conjunct of D has. So D is unsatisfiable exactly when $\neg D$, and so K, is a tautology, which by (a) happens exactly when every conjunct of D has such a pair. That is (b) again.

The proposition trades one kind of work for another. Checking a CNF for validity is now a glance, but getting the CNF in the first place may take the 2^n rows of the table route or the explosion of the rewriting route. Nothing is free: the work has moved, not vanished.

Example. Validity at a glance

Decide whether each formula is a tautology, without a table.

(a) $(\neg p \vee q \vee p) \wedge (r \vee \neg q \vee s \vee q) \wedge (\neg s \vee \neg r \vee s) \wedge (p \vee \neg r \vee r \vee q)$.

(b) The same formula with the third clause changed to $\neg s \vee \neg r \vee q$.

Solution. *(a)* Each clause has a pair: $\neg p, p$ in the first; $\neg q, q$ in the second; $\neg s, s$ in the third; $\neg r, r$ in the fourth. By Proposition 4.8.3 (a), it is a tautology. A table would have needed 16 rows.

(b) The new third clause $\neg s \vee \neg r \vee q$ has the atoms s, r, q, each once, so it has no pair, and the formula is not a tautology. The proof of Proposition 4.8.3 names a refuting row: $s = \mathrm{T}$, $r = \mathrm{T}$, $q = \mathrm{F}$, and $p = \mathrm{T}$, like every atom not in the clause; in fact either value of p works. In that row the third clause is F, so the conjunction is F.

4.8.6 Adequate sets of connectives

Our formulas use five connectives and $\bot$. Chapter 2 already showed that some of them can be traded for others: an implication for a disjunction (Proposition 2.5.6), a biconditional for two implications (Proposition 2.4.1). How far can this go? Before we can answer, we need to say what a set of connectives must be able to "say".

Every formula whose atoms are among $p_1, \ldots, p_n$ has a truth table: a value for each of the 2^n rows. Turn this around, and ask whether **every** table is the table of some formula.

Definition 4.8.4. Truth Function

Let $n \geq 1$. A **truth function** with n inputs assigns to **each** list $(a_1, \ldots, a_n)$ of truth values (each a_i is T or F) a truth value $f(a_1, \ldots, a_n)$. A formula φ **expresses** f if

$$v(\varphi) = f(v(p_1), \ldots, v(p_n)) \quad \text{for every valuation } v.$$

In words: a truth function is a truth table with n input columns and one output column, with no formula attached. There are 2^n lists of inputs and two choices of output for each, so there are 2^{2^n} truth functions with n inputs: 4 with one input and 16 with two. Examples: the **negation** function with one input, $\mathrm{T} \mapsto \mathrm{F}$, $\mathrm{F} \mapsto \mathrm{T}$, expressed by $\neg p_1$; the two-input function that is T only at (T, T), expressed by $p_1 \wedge p_2$; the three-input **majority** function, T when at least two inputs are T, expressed by $(p_1 \wedge p_2) \vee (p_1 \wedge p_3) \vee (p_2 \wedge p_3)$. A degenerate case: the one-input function that is always T is expressed by $p_1 \vee \neg p_1$, and also by $p_2 \to p_2$, which does not mention p_1 at all. The definition allows that.

We say a formula is **in** a set S of connectives if every connective occurring in it belongs to S, and $\bot$ occurs in it only if $\bot$ belongs to S (for this purpose $\bot$ counts as a connective with no inputs).

Definition 4.8.5. Adequate Set of Connectives

A set S of connectives is **adequate** if **every** truth function, with **any** number $n \geq 1$ of inputs, is expressed by some formula in S.

A set containing an adequate set is adequate, since a formula in the smaller set is also in the larger one. The theorem below shows that $\{\neg, \wedge, \vee\}$ is adequate, so the full set $\{\neg, \wedge, \vee, \to, \leftrightarrow, \bot\}$

is too. The question is which smaller sets are.

Theorem 4.8.4. Adequate Sets

(a) The sets $\{\neg, \wedge, \vee\}$, $\{\neg, \wedge\}$, $\{\neg, \vee\}$ and $\{\rightarrow, \bot\}$ are adequate.

(b) The set $\{\wedge, \vee, \rightarrow\}$ is not adequate: no formula in it expresses the negation function.

Idea. For (a), the construction of Theorem 4.8.1 works for any table, not only for the table of a formula, and its output uses only $\neg$, $\wedge$, $\vee$. To shrink the set further, rewrite each missing connective in terms of the others, for instance $\alpha \vee \beta$ as $\neg(\neg\alpha \wedge \neg\beta)$, recursively through the formula. For (b), look at the row where every atom is true: $\wedge$, $\vee$ and $\rightarrow$ all give T on T inputs, so every formula in them is T there, while negation must give F.

Proof. (a) The set $\{\neg, \wedge, \vee\}$. Let f be a truth function with $n \geq 1$ inputs. For each of the 2^n lists of inputs, fix a valuation taking those values at $p_1, \ldots, p_n$, and let $v_1, \ldots, v_k$ be those fixed for the lists where f is T. If $k \geq 1$, let $D \coloneqq C_{v_1} \vee \cdots \vee C_{v_k}$, with the conjuncts C_v formed from $p_1, \ldots, p_n$; if $k = 0$, let $D \coloneqq p_1 \wedge \neg p_1$. By Eq. 4.8.1 and Eq. 4.8.2, a valuation w makes D true exactly when $(w(p_1), \ldots, w(p_n))$ is a list where f is T, that is, when $f(w(p_1), \ldots, w(p_n)) = \mathrm{T}$. So D expresses f, and D is in $\{\neg, \wedge, \vee\}$.

The other three sets. By Theorem 4.3.4, the following clauses define three functions t_1, t_2, t_3 on all formulas. Each has $t_i(p) \coloneqq p$ for every atom p, and $t_i(\bot) \coloneqq \bot$, $t_i(\alpha \rightarrow \beta) \coloneqq t_i(\alpha) \rightarrow t_i(\beta)$ and $t_i(\alpha \leftrightarrow \beta) \coloneqq t_i(\alpha) \leftrightarrow t_i(\beta)$ (these last clauses are never used below). The other clauses rewrite the connectives $\neg$, $\wedge$, $\vee$:

- t_1, into $\{\neg, \wedge\}$: $t_1(\neg\alpha) \coloneqq \neg t_1(\alpha)$, $t_1(\alpha \wedge \beta) \coloneqq t_1(\alpha) \wedge t_1(\beta)$, and $t_1(\alpha \vee \beta) \coloneqq \neg(\neg t_1(\alpha) \wedge \neg t_1(\beta))$;
- t_2, into $\{\neg, \vee\}$: $t_2(\neg\alpha) \coloneqq \neg t_2(\alpha)$, $t_2(\alpha \wedge \beta) \coloneqq \neg(\neg t_2(\alpha) \vee \neg t_2(\beta))$, and $t_2(\alpha \vee \beta) \coloneqq t_2(\alpha) \vee t_2(\beta)$;
- t_3, into $\{\rightarrow, \bot\}$: $t_3(\neg\alpha) \coloneqq t_3(\alpha) \rightarrow \bot$, $t_3(\alpha \wedge \beta) \coloneqq (t_3(\alpha) \rightarrow (t_3(\beta) \rightarrow \bot)) \rightarrow \bot$, and $t_3(\alpha \vee \beta) \coloneqq (t_3(\alpha) \rightarrow \bot) \rightarrow t_3(\beta)$.

Claim. Let $S_1 = \{\neg, \wedge\}$, $S_2 = \{\neg, \vee\}$, $S_3 = \{\rightarrow, \bot\}$. For every formula φ in $\{\neg, \wedge, \vee\}$ and each i, the formula $t_i(\varphi)$ is in S_i, and $v(t_i(\varphi)) = v(\varphi)$ for every valuation v.

Proof of Claim. By structural induction (Theorem 4.3.1) on φ, with the property "if φ is in $\{\neg, \wedge, \vee\}$, then for each i, $t_i(\varphi)$ is in S_i and has the same value as φ under every valuation". The immediate parts of a formula in $\{\neg, \wedge, \vee\}$ are in $\{\neg, \wedge, \vee\}$, so the induction hypothesis applies to them. For $\bot$, $\alpha \rightarrow \beta$ and $\alpha \leftrightarrow \beta$ the property holds vacuously, since these formulas are not in $\{\neg, \wedge, \vee\}$. For an atom, $t_i(p) = p$. In the other cases, write $a = v(t_i(\alpha)) = v(\alpha)$ and $b = v(t_i(\beta)) = v(\beta)$, by the induction hypothesis. Each clause for t_i above uses only the connectives of S_i, applied to formulas in S_i, so $t_i(\varphi)$ is in S_i. For the values, use Definition 4.4.2, and that $\chi \rightarrow \bot$ is true exactly when χ is false ($\bot$ is always F):

- $\neg t_i(\alpha)$ and $t_3(\alpha) \rightarrow \bot$ are true exactly when $a = \mathrm{F}$, as is $\neg\alpha$.
- $\neg(\neg t_2(\alpha) \vee \neg t_2(\beta))$ is true exactly when neither $\neg t_2(\alpha)$ nor $\neg t_2(\beta)$ is, that is, when $a = b = \mathrm{T}$. And $(t_3(\alpha) \rightarrow (t_3(\beta) \rightarrow \bot)) \rightarrow \bot$ is true exactly when $t_3(\alpha) \rightarrow (t_3(\beta) \rightarrow \bot)$ is false, that is, when $a = \mathrm{T}$ and $t_3(\beta) \rightarrow \bot$ is false, that is, when $a = b = \mathrm{T}$. Both agree with $\alpha \wedge \beta$.
- $\neg(\neg t_1(\alpha) \wedge \neg t_1(\beta))$ is true exactly when a and b are not both F. And $(t_3(\alpha) \rightarrow \bot) \rightarrow$

$t_3(\beta)$ is true exactly when $t_3(\alpha) \to \bot$ is false or $b = \mathrm{T}$, that is, when $a = \mathrm{T}$ or $b = \mathrm{T}$. Both agree with $\alpha \vee \beta$.

The remaining clauses, for $t_1(\alpha \wedge \beta)$ and $t_2(\alpha \vee \beta)$, keep the connective and have the right value by the induction hypothesis.

■

Now let f be any truth function, and D the formula in $\{\neg, \wedge, \vee\}$ that expresses it. By the Claim, $t_i(D)$ is in S_i and $v(t_i(D)) = v(D) = f(v(p_1), \ldots, v(p_n))$ for every valuation v. So each S_i is adequate.

(b) Let v_{T} be the valuation with $v_{\mathrm{T}}(p) = \mathrm{T}$ for every atom p. We show by structural induction that every formula in $\{\wedge, \vee, \to\}$ is true under v_{T}; the property is "if φ is in $\{\wedge, \vee, \to\}$, then $v_{\mathrm{T}}(\varphi) = \mathrm{T}$". For an atom, $v_{\mathrm{T}}(p) = \mathrm{T}$. The formulas $\bot$, $\neg\alpha$ and $\alpha \leftrightarrow \beta$ are not in the set, so for them the property holds vacuously. If φ is $\alpha \wedge \beta$, $\alpha \vee \beta$ or $\alpha \to \beta$ and is in the set, then so are α and β, and by the induction hypothesis $v_{\mathrm{T}}(\alpha) = v_{\mathrm{T}}(\beta) = \mathrm{T}$. By Definition 4.4.2, $v_{\mathrm{T}}(\alpha \wedge \beta) = \mathrm{T}$ and $v_{\mathrm{T}}(\alpha \vee \beta) = \mathrm{T}$; and $v_{\mathrm{T}}(\alpha \to \beta) = \mathrm{T}$ because its conclusion β is true. By Theorem 4.3.1, every formula in the set is true under v_{T}.

Now suppose some φ in $\{\wedge, \vee, \to\}$ expressed the negation function f. Then $v_{\mathrm{T}}(\varphi) = f(v_{\mathrm{T}}(p_1)) = f(\mathrm{T}) = \mathrm{F}$, contradicting $v_{\mathrm{T}}(\varphi) = \mathrm{T}$. So no such φ exists, and the set is not adequate. This proves the theorem.

■

In the implication case of (b), the reason is that the **conclusion** β is true. That is where a careless argument goes wrong, and Exercise (C2: Find the error) asked you to find a "proof" that goes wrong there.

The five connectives were chosen because each matches a word mathematicians use, not because they are needed: two suffice, and Exercise (C1: One connective is enough) shows that one well-chosen connective suffices.

Warning. The $\bot$ in $\{\to, \bot\}$ is essential. Without it, $\{\to\}$ is not adequate: every formula in $\{\to\}$ is also in $\{\wedge, \vee, \to\}$, so by (b) of Theorem 4.8.4 none expresses negation. Implication alone cannot say "false"; implication together with a formula that is always false can.

Quick check. Use the translation t_3 to express $p \wedge q$ with $\to$ and $\bot$ only, and check your formula in the row $p = \mathrm{T}$, $q = \mathrm{F}$.

Solution. $t_3(p \wedge q) = (p \to (q \to \bot)) \to \bot$. In the row $p = \mathrm{T}$, $q = \mathrm{F}$: $q \to \bot$ is $\mathrm{F} \to \mathrm{F}$, so T; then $p \to (q \to \bot)$ is T; then the whole is $\mathrm{T} \to \bot$, so F, which agrees with $p \wedge q$.

4.8.7 Exercises

4.8.7.1 A. Check your understanding

Exercise. A1

(a) Which of $\neg p$, $\neg\neg q$, $\bot$, r and $\neg(p \vee q)$ are literals?

(b) Classify each as a clause, a conjunct, both, or neither: $p \vee \neg q$; $\neg r$; $p \wedge (q \vee r)$.

(c) Is $(p \vee q) \wedge (\neg p \vee p \vee r)$ a tautology? Is $(q \vee \neg q) \wedge (\neg r \vee p \vee r)$? Give a refuting row where the answer is no.

(d) Give a formula in DNF that is not satisfiable.

(e) True or false: every formula in CNF with only one clause is also in DNF.

Solution. *(a)* $\neg p$ and r. $\neg\neg q$ has two negations, $\bot$ is not an atom, and $\neg(p \vee q)$ negates a disjunction.

(b) $p \vee \neg q$ is a clause. $\neg r$ is both (one literal). $p \wedge (q \vee r)$ is neither: its part $q \vee r$ is not a literal.

(c) The first is not a tautology: its first clause $p \vee q$ has no pair, and the row $p = q = \mathrm{F}$ (any r) makes it false, hence the whole conjunction. The second is: each clause has a pair ($q, \neg q$ and $r, \neg r$), by Proposition 4.8.3 (a).

(d) $p \wedge \neg p$: its one conjunct contains a pair, so by Proposition 4.8.3 (b) it is not satisfiable.

(e) True. A single clause $\ell_1 \vee \cdots \vee \ell_m$ is a disjunction of m conjuncts of one literal each.

4.8.7.2 B. Practice

Exercise. B1: Normal forms of three formulas

Find a DNF and a CNF of each formula, by either route, and check one of them in two rows.

(a) $p \leftrightarrow \neg q$.

(b) $(p \wedge q) \rightarrow r$.

(c) $\neg(p \vee (q \rightarrow r))$.

Solution. *(a)* By the table: $p \leftrightarrow \neg q$ is true when p and q have opposite values, so its T-rows are T, F and F, T, and its F-rows are T, T and F, F. DNF: $(p \wedge \neg q) \vee (\neg p \wedge q)$, as in Example (Exactly one of two). CNF, one clause per F-row from the opposite literals: the row T, T gives $\neg p \vee \neg q$ and the row F, F gives $p \vee q$, so $(\neg p \vee \neg q) \wedge (p \vee q)$. Check the CNF: in the row T, F both clauses are true, and so is $p \leftrightarrow \neg q$; in the row T, T the first clause is false, and so is $p \leftrightarrow \neg q$.

(b) By rewriting: $(p \wedge q) \rightarrow r \equiv \neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$. This is a single clause, so it is a CNF, and it is also a DNF with three one-literal conjuncts. Check: in the row T, T, F it is false (every literal false), as is $(p \wedge q) \rightarrow r$; in the row F, T, F it is true, as is the original (its hypothesis is false).

(c) By rewriting: $\neg(p \vee (q \rightarrow r)) \equiv \neg p \wedge \neg(q \rightarrow r) \equiv \neg p \wedge \neg(\neg q \vee r) \equiv \neg p \wedge q \wedge \neg r$. This single conjunct is a DNF, and it is also a CNF with three one-literal clauses. Check: in the row F, T, F it is true, and so is the original ($q \rightarrow r$ is false and p is false); in the row F, T, T it is false, and so is the original ($q \rightarrow r$ is true).

Exercise. B2: Two connectives from two

Find formulas in $\{\neg, \wedge\}$, with atoms p and q, equivalent to $p \vee q$ and to $p \leftrightarrow q$. Check the second by a truth table.

Solution. $p \vee q \equiv \neg(\neg p \wedge \neg q)$: this is the translation t_1 in the proof of Theorem 4.8.4, and the right side is F exactly when $\neg p$ and $\neg q$ are both true, that is, when $p \vee q$ is F.

For the biconditional, $p \leftrightarrow q$ says $p \to q$ and $q \to p$, and $p \to q$ says "not p without q", that is, $\neg(p \wedge \neg q)$. So $p \leftrightarrow q \equiv \neg(p \wedge \neg q) \wedge \neg(\neg p \wedge q)$. The table:

p	q	$\neg(p \wedge \neg q)$	$\neg(\neg p \wedge q)$	$\neg(p \wedge \neg q) \wedge \neg(\neg p \wedge q)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

The last two columns agree in every row.

Exercise. B3: Odd parity

Give an example of a formula in DNF, with atoms p, q, r, that is true in exactly the rows where an **odd** number of p, q, r are true. List the properties your answer must have, and check it against them.

Solution. The rows with an odd number of T's are those with three T's, T T T, and those with one, T F F, F T F, and F F T. One conjunct per row, as in Theorem 4.8.1:

$$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r).$$

The properties an answer must have: (i) it is in DNF (a disjunction of conjunctions of literals); (ii) its atoms are among p, q, r; (iii) it is T in the four rows T T T, T F F, F T F, F F T; (iv) it is F in the other four, T T F, T F T, F T T, F F F. Check: (i) and (ii) hold by inspection. By Eq. 4.8.2, each conjunct is T in its own row only, so the disjunction is T exactly in the four rows listed, which gives (iii) and (iv).

A different correct answer, for instance the same conjuncts in another order, or with the literals of a conjunct reordered, passes the same checks. No DNF with fewer conjuncts works: a conjunct omitting an atom, say $p \wedge q$, would be T in two rows that differ at that atom, and changing one atom always changes the parity.

4.8.7.3 C. Going deeper

Exercise. C1: One connective is enough

Add to the language two binary connectives: $\alpha \mid \beta$ ("not both", NAND), which is F exactly when α and β are both T, and $\alpha \downarrow \beta$ ("neither", NOR), which is T exactly when α and β are both F.

(a) Prove that $\{|\}$ is adequate.

(b) Prove that $\{\downarrow\}$ is adequate.

Solution. *(a)* First, two equivalences, checked from the table of $|$: $\alpha \,|\, \alpha$ is F exactly when α is T, so $\alpha \,|\, \alpha \equiv \neg\alpha$; and $(\alpha \,|\, \beta) \,|\, (\alpha \,|\, \beta) \equiv \neg(\alpha \,|\, \beta)$, which is T exactly when α and β are both T, so it is equivalent to $\alpha \wedge \beta$. Now define, by recursion on formulas as in Theorem 4.3.4, $s(p) \coloneqq p$, $s(\neg\alpha) \coloneqq s(\alpha) \,|\, s(\alpha)$, $s(\alpha \wedge \beta) \coloneqq (s(\alpha) \,|\, s(\beta)) \,|\, (s(\alpha) \,|\, s(\beta))$ (the other clauses do not matter). By structural induction, exactly as in the Claim in the proof of Theorem 4.8.4, every formula φ in $\{\neg, \wedge\}$ has $s(\varphi)$ in $\{|\}$ and $v(s(\varphi)) = v(\varphi)$ for every v: the atom case is immediate, and the two equivalences above give the $\neg$ and $\wedge$ cases from the induction hypothesis. Given a truth function f, Theorem 4.8.4 (a) gives a formula φ in $\{\neg, \wedge\}$ expressing it, and then $s(\varphi)$ expresses f and is in $\{|\}$.

(b) The same, with $\{\neg, \vee\}$: $\alpha \downarrow \alpha$ is T exactly when α is F, so $\alpha \downarrow \alpha \equiv \neg\alpha$; and $(\alpha \downarrow \beta) \downarrow (\alpha \downarrow \beta) \equiv \neg(\alpha \downarrow \beta) \equiv \alpha \vee \beta$. The recursive translation $s(\neg\alpha) \coloneqq s(\alpha) \downarrow s(\alpha)$, $s(\alpha \vee \beta) \coloneqq (s(\alpha) \downarrow s(\beta)) \downarrow (s(\alpha) \downarrow s(\beta))$ turns a formula in $\{\neg, \vee\}$ expressing f into one in $\{\downarrow\}$.

(Recursion and structural induction for the language with the new connectives hold by the same proofs as Theorem 4.3.4 and Theorem 4.3.1, with one more formation clause.)

Marking scheme. (10 points)

- 3: (a) the two equivalences $\alpha \,|\, \alpha \equiv \neg\alpha$ and $(\alpha \,|\, \beta) \,|\, (\alpha \,|\, \beta) \equiv \alpha \wedge \beta$, read off the table of $|$.
- 3: (a) the recursive translation and the structural induction showing that it keeps the value under every valuation.
- 1: (a) the conclusion, via Theorem 4.8.4 (a).
- 3: (b) the same for $\downarrow$, with $\{\neg, \vee\}$.

Fatal errors (0 for the step): checking the translation on examples only, with no induction.

Exercise. C2: Two sets that are not adequate

(a) Prove that $\{\wedge, \vee, \rightarrow, \leftrightarrow\}$ is not adequate.

(b) Add to the language a three-place connective $\mathsf{M}(\alpha, \beta, \gamma)$, **majority**, which is T exactly when at least two of α, β, γ are T. Prove that $\{\neg, \mathsf{M}\}$ is not adequate.

Solution (Hint 1). For (b): compare each row with the row in which every atom's value is flipped.

Solution (Hint 2). For (b): show that every formula in $\neg$ and M takes opposite values in a row and in its flipped row.

Solution. *(a)* Let v_{T} make every atom true. Every formula in $\{\wedge, \vee, \rightarrow, \leftrightarrow\}$ is true under v_{T}, by the structural induction in the proof of Theorem 4.8.4 (b) with one more case: if

$v_{\mathrm{T}}(\alpha) = v_{\mathrm{T}}(\beta) = \mathrm{T}$, then $v_{\mathrm{T}}(\alpha \leftrightarrow \beta) = \mathrm{T}$, since the two values are equal. As there, no such formula expresses negation, which must give F at the input T.

(b) For a valuation v, let $\bar{v}$ be the **flipped** valuation: $\bar{v}(p)$ is the opposite of $v(p)$, for every atom p. We show by structural induction that every formula φ in $\{\neg, \mathsf{M}\}$ satisfies: for every valuation v, $\bar{v}(\varphi)$ is the opposite of $v(\varphi)$. For an atom this is the definition of $\bar{v}$. If $\bar{v}(\alpha)$ is the opposite of $v(\alpha)$, then $\bar{v}(\neg\alpha)$, the opposite of $\bar{v}(\alpha)$, is $v(\alpha)$, which is the opposite of $v(\neg\alpha)$. If the property holds for α, β, γ, then flipping v flips all three values; at least two of them are T after the flip exactly when at least two were F before, that is, exactly when at most one was T, that is, exactly when $\mathsf{M}(\alpha, \beta, \gamma)$ was F. So $\bar{v}(\mathsf{M}(\alpha, \beta, \gamma))$ is the opposite of $v(\mathsf{M}(\alpha, \beta, \gamma))$. The other connectives and $\bot$ do not occur in formulas of $\{\neg, \mathsf{M}\}$, so those cases are vacuous. (Structural induction for the extended language holds by the proof of Theorem 4.3.1, with one more formation clause.)

Now the two-input function of $p_1 \wedge p_2$ is F at (T, F) and at the flipped input (F, T). If a formula φ in $\{\neg, \mathsf{M}\}$ expressed it, then for a valuation v with $v(p_1) = \mathrm{T}$, $v(p_2) = \mathrm{F}$ we would have $v(\varphi) = \mathrm{F}$ and $\bar{v}(\varphi) = \mathrm{F}$, since $\bar{v}(p_1) = \mathrm{F}$ and $\bar{v}(p_2) = \mathrm{T}$. But these values must be opposite. So no formula in $\{\neg, \mathsf{M}\}$ expresses it, and the set is not adequate.

Marking scheme. (10 points)

- 3: (a) states the all-T valuation and runs the induction, with the $\leftrightarrow$ case and the vacuous cases ($\neg$, $\bot$) named.
- 1: (a) names a truth function that is not expressed (negation), with the reason.
- 4: (b) states the flipped-row property for **every** formula in $\{\neg, \mathsf{M}\}$ and proves it by structural induction, including the M case.
- 2: (b) exhibits a truth function that is F on a row and on its flipped row, and concludes.

Fatal errors (0 for the step): in (b), checking the flipped-row property on a few formulas instead of proving it for all; in (a), an implication case justified by the hypothesis being true rather than by the conclusion.

Exercise. C3: Exit check: deciding sequents

For each sequent, decide whether it holds. If it does, justify this by a derivation, or by a truth table and the theorem that turns the table into a derivation; if it does not, give a refuting row and say which theorem turns the row into a proof that no derivation exists.

(a) $(p \to q) \vee (p \to r) \vdash p \to (q \vee r)$.

(b) $(p \to q) \to r \vdash p \to r$.

(c) $p \vee (q \wedge r) \vdash (p \vee q) \wedge r$.

(d) $(p \vee q) \wedge \neg p \vdash q \vee r$.

Solution. (a) Derivable. The goal is an implication: a box on p, with the goal $q \vee r$; the premise is a disjunction, used by $\vee$E inside the box; in each case $\to$E with p gives one side, and $\vee$I gives the goal.

1	$(p \to q) \vee (p \to r)$	
2	$\quad p$	asm, for $\to$I
3	$\quad\quad p \to q$	asm, for $\vee$E
4	$\quad\quad q$	$\to$E 3, 2
5	$\quad\quad q \vee r$	$\vee$I 4
6	$\quad\quad p \to r$	asm, for $\vee$E
7	$\quad\quad r$	$\to$E 6, 2
8	$\quad\quad q \vee r$	$\vee$I 7
9	$\quad q \vee r$	$\vee$E 1, 3–5, 6–8
10	$p \to (q \vee r)$	$\to$I 2–9

(b) Not derivable. In the row $p = \mathrm{T}$, $q = \mathrm{F}$, $r = \mathrm{F}$, the premise is true ($p \to q$ is F, so $(p \to q) \to r$ is T) and the conclusion $p \to r$ is F. By Corollary 4.6.3, $(p \to q) \to r \nvdash p \to r$.

(c) Not derivable. In the row $p = \mathrm{T}$, $q = \mathrm{T}$, $r = \mathrm{F}$ (the value of q does not matter), the premise is true because p is, and the conclusion is false because r is. By Corollary 4.6.3, the sequent fails.

(d) Derivable. Split the premise by $\wedge$E, then use $p \vee q$ by $\vee$E: the case p clashes with $\neg p$ and closes by $\bot$E; the case q gives the goal by $\vee$I.

1	$(p \vee q) \wedge \neg p$	
2	$p \vee q$	$\wedge$E 1
3	$\neg p$	$\wedge$E 1
4	$\quad p$	asm, for $\vee$E
5	$\quad \bot$	$\neg$E 4, 3
6	$\quad q \vee r$	$\bot$E 5
7	$\quad q$	asm, for $\vee$E
8	$\quad q \vee r$	$\vee$I 7
9	$q \vee r$	$\vee$E 2, 4–6, 7–8

Neither derivation uses RAA. As the problem allows, (a) and (d) may instead be justified by a table (or backwards from F) showing the entailment, followed by Theorem 4.7.5.

Marking scheme. (10 points; 2.5 per sequent)

- 1 per sequent: the correct verdict.
- 1.5 per sequent: its justification: a derivation in which every line is justified and every citation available, or a table check with Theorem 4.7.5; or a row that makes **every** premise true and the conclusion false, checked, with Corollary 4.6.3 (or Theorem 4.6.2) cited.

Fatal errors (0 for the sequent's justification): a "refuting row" that makes some premise

false; a derivation citing a line that is not available.

Exercise. C4: Exit check: normal forms

Put $(p \to q) \to (q \wedge r)$ into DNF and into CNF, by either route, and check one of your answers against the truth table.

Solution. By rewriting. Eliminate the outer implication and then the inner one, and push the negation in:

$$\begin{aligned}(p \to q) \to (q \wedge r) &\equiv \neg(p \to q) \vee (q \wedge r)\\ &\equiv \neg(\neg p \vee q) \vee (q \wedge r)\\ &\equiv (p \wedge \neg q) \vee (q \wedge r).\end{aligned}$$

The last line is a DNF. Here $\neg(p \to q) \equiv p \wedge \neg q$, since an implication is false exactly when its hypothesis is true and its conclusion false. For a CNF, distribute:

$$(p \wedge \neg q) \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \wedge (\neg q \vee q) \wedge (\neg q \vee r),$$

and drop the clause $\neg q \vee q$, which is true in every row: the CNF is $(p \vee q) \wedge (p \vee r) \wedge (\neg q \vee r)$. (The clause $p \vee r$ is redundant too: $(p \vee q) \wedge (\neg q \vee r)$ is an equivalent CNF, as splitting on q shows. Either is correct.)

The check. The formula is false exactly when $p \to q$ holds and $q \wedge r$ fails. Its table:

p	q	r	$p \to q$	$q \wedge r$	$(p \to q) \to (q \wedge r)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	T	F	F

The T-rows are T T T, T F T, T F F, and F T T. The DNF $(p \wedge \neg q) \vee (q \wedge r)$ is true in exactly these: $p \wedge \neg q$ in T F T and T F F, and $q \wedge r$ in T T T and F T T. In each of the four F-rows the CNF has a false clause: $\neg q \vee r$ in T T F and F T F, $p \vee q$ in F F T and F F F.

By the table route, the DNF has four conjuncts, one per T-row, and the CNF four clauses, one per F-row, for instance $\neg p \vee \neg q \vee r$ for the row T T F. Any correct DNF and CNF earns full marks; a CNF is not unique.

Marking scheme. (10 points)

- 2: eliminates $\to$ correctly, with $\neg(p \to q) \equiv p \wedge \neg q$ and its reason.
- 3: a correct DNF.
- 3: a correct CNF, by distribution carried out in full or by one clause per F-row.
- 2: the check against the table, covering at least the four F-rows (or the four T-rows).

Fatal errors (0 for the step): $\neg(p \to q)$ rewritten as $\neg p \to \neg q$; a "CNF" with a conjunction inside a clause.

4.8.7.4 R. Review

Exercise. R1

Find a derivation of $p \lor q,\ \neg(p \land r),\ r \vdash q$.

Solution. The method of Section 3.1 (Example (Proof by cases, found by the method) shows it on a proof by cases). The goal q is an atom, so no introduction rule applies; the premise $p \lor q$ is a disjunction, so we use it by $\lor$E, with the goal q in both boxes. The case q is the goal. In the case p, nothing gives q directly, so we look for a clash: p and the premise r build $p \land r$ by $\land$I, which clashes with $\neg(p \land r)$, and $\bot$E closes the impossible case (Example (Manufacturing the other half of a clash)).

1	$p \lor q$	
2	$\neg(p \land r)$	
3	r	
4	$\quad p$	asm, for $\lor$E
5	$\quad p \land r$	$\land$I 4, 3
6	$\quad \bot$	$\neg$E 5, 2
7	$\quad q$	$\bot$E 6
8	$\quad q$	asm, for $\lor$E
9	q	$\lor$E 1, 4–7, 8–8

The second box is its assumption alone, which is already the goal. No RAA is used.

Exercise. R2

Let m and n be integers, and consider the argument: "$m + n$ is even only if m and n are both even or both odd. m is even and n is odd. Therefore $m + n$ is not even." Translate it into a sequent, using atoms e: "$m + n$ is even", a: "m is even", b: "n is even", and decide whether the premises entail the conclusion.

Solution. "P only if Q" is $P \to Q$ (Section 1.1, and Exercise (C1: "Only if" and "unless")). "Both odd" is $\neg a \land \neg b$, since an integer is odd exactly when it is not even. So the sequent is

$$e \to ((a \land b) \lor (\neg a \land \neg b)),\quad a \land \neg b \models \neg e.$$

Work backwards from F. A refuting row needs $\neg e$ false, so e = T, and the second premise true, so a = T, b = F. In that row $a \land b$ and $\neg a \land \neg b$ are both F, so the first premise is T $\to$ F, which is F. So no row makes both premises true and the conclusion false: the entailment holds. By Theorem 4.7.5, the sequent is also derivable. (The eight-row table gives the same

answer; working backwards found the only candidate row at once.)

Exercise. R3

Is $(p \wedge (q \vee r)) \to ((p \wedge q) \vee r)$ a tautology? Is it derivable without RAA?

Solution. A tautology. Work backwards from F (Section 4.4). The formula is false only if $p \wedge (q \vee r)$ is true and $(p \wedge q) \vee r$ is false. The second needs r false and $p \wedge q$ false; the first needs p true and $q \vee r$ true, so with r false, q true. But then $p \wedge q$ is true. No row makes the formula false, so it is a tautology (Definition 4.5.1).

Derivable without RAA. Completeness (Theorem 4.7.5) says it is derivable, but not how, and its proof uses LEM in Lemma 4.7.4, so it does not answer the second question. A search does: the goal is an implication, so a box on $p \wedge (q \vee r)$; split it by $\wedge$E and use $q \vee r$ by $\vee$E.

1	$p \wedge (q \vee r)$	asm, for $\to$I
2	p	$\wedge$E 1
3	$q \vee r$	$\wedge$E 1
4	q	asm, for $\vee$E
5	$p \wedge q$	$\wedge$I 2, 4
6	$(p \wedge q) \vee r$	$\vee$I 5
7	r	asm, for $\vee$E
8	$(p \wedge q) \vee r$	$\vee$I 7
9	$(p \wedge q) \vee r$	$\vee$E 3, 4–6, 7–8
10	$(p \wedge (q \vee r)) \to ((p \wedge q) \vee r)$	$\to$I 1–9

No RAA is used: the boxes close by $\vee$E and $\to$I. The tools: truth values (Definition 4.4.2) for the first question, the search of Chapter 3 for the second. The look-alike tool, completeness, answers only "derivable with all the rules".